



CS 540 Introduction to Artificial Intelligence

Games I

University of Wisconsin-Madison
Spring 2025

Announcements

- **Homework:**
 - HW8 due on **Wednesday 16th at 11:59 PM**

- Class roadmap:

Games – Part I

Games – Part II

Reinforcement Learning

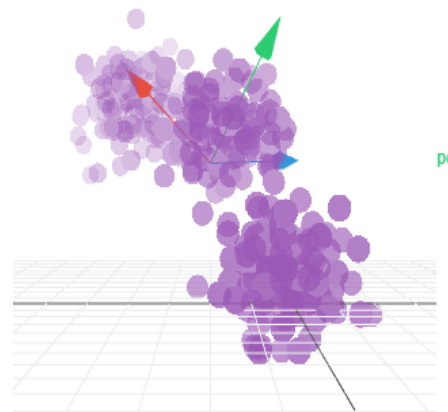
Outline

- Introduction to game theory
 - Properties of games, mathematical formulation
- Simultaneous-Move Games
 - Normal form, strategies, dominance, Nash equilibrium

So Far in The Course

We looked at techniques:

- **Unsupervised:** See data, do something with it. Unstructured.
- **Supervised:** Train a model to make predictions. More structure (labels).
- **Planning and Games:** Much more structure.



Victor Powell



indoor

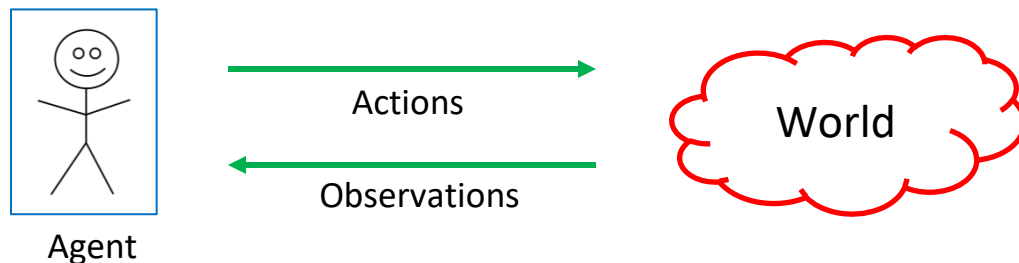


outdoor



More General Model

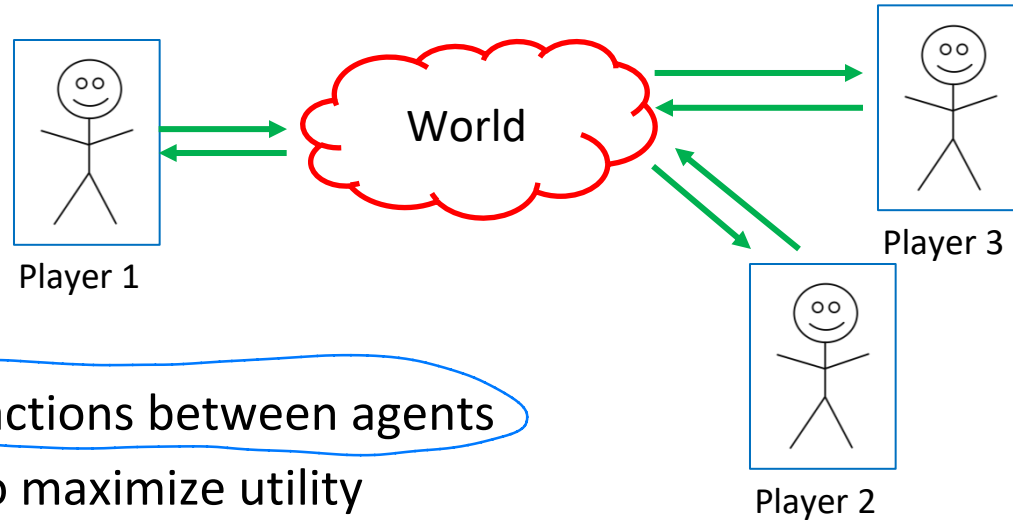
Suppose we have an agent interacting with the world



- Agent receives a reward based on state of the world
 - Goal: maximize reward / utility (\$\$\$)
 - Note: now **data** consists of actions, observations, and rewards
 - Setup for decision theory, reinforcement learning, planning

Games: Multiple Agents

Games setup: multiple agents



- Now: interactions between agents
- Still want to maximize utility
- Requires **strategic** decision making.

Modeling Games: Properties

Let's work through **properties** of games

- **Number** of agents/players
- Action space: **finite** or **infinite**
- **Deterministic** or **random**
- **Zero**-sum or **general**-sum
- **Sequential** or **simultaneous moves**

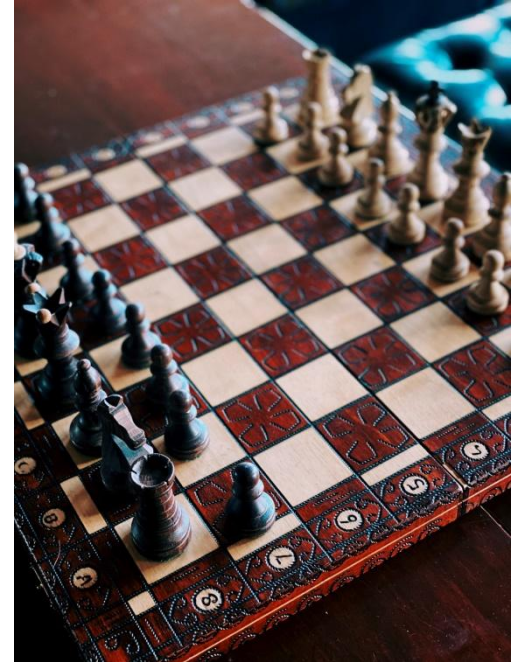


Wiki

Property 1: Number of players

1 or more players

- Usually interested in ≥ 2 players
- Typically a finite number of players



Property 2: Action Space

Action space: set of possible actions an agent can choose from.

Can be finite or infinite.

Examples:

- Rock-paper-scissors
- Tennis

Property 3: **Deterministic** or **Random**

- Is there **chance** in the game?
 - Poker → *There is randomness existed in this*
 - Chess
 - Scrabble



Property 4: Sum of payoffs

- Two basic types: zero sum vs. general sum.
- Zero sum: one player's win is the other's loss
 - Pure competition.
 - Example: rock-paper-scissors
- General sum
 - Example: driving to work, prisoner's dilemma

Property 5: **Sequential** or **Simultaneous Moves**

- Simultaneous: all players take action at the same time
- Sequential: take turns
 - But payoff is often only revealed at end of game

Quiz break:

Give the properties of the game shown on the right:

- Number of players? 7
- Deterministic or stochastic? sto
- Sum of pay-offs? zero-sum
- Finite or infinite action-space? ?
- Sequential or simultaneous?

Sequential



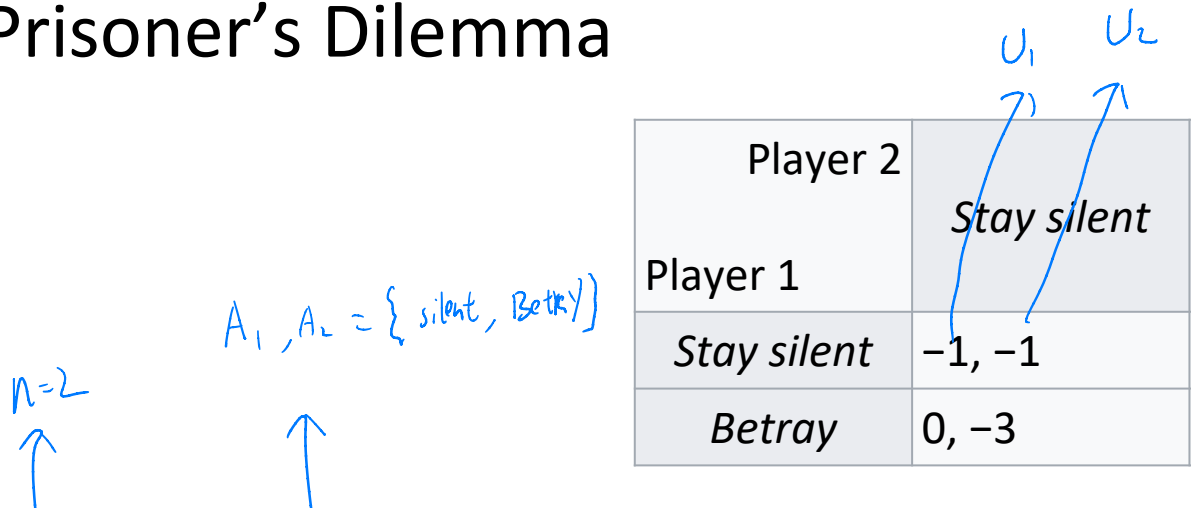
Normal Form Game

Mathematical description of simultaneous games.

- n players $\{1, 2, \dots, n\}$
- Player i chooses strategy a_i from action space A_i .
single action (pointing to a_i) *All the actions they can do* (pointing to A_i)
- Strategy profile: $a = (a_1, a_2, \dots, a_n)$ → *The actions each player takes (vector)*
- Player i gets rewards $u_i(a)$
- ★ — Note: reward depends on other players!
As a function of all actions from all players (pointing to a)
- We consider the simple case where all reward functions are common knowledge.
Rewards are known, no uncertainty (pointing to the underlined phrase)

Example of Normal Form Game

Ex: Prisoner's Dilemma



Player 2		
Player 1	<i>Stay silent</i>	<i>Betray</i>
	<i>Stay silent</i> -1, -1	-3, 0
	<i>Betray</i> 0, -3	-2, -2

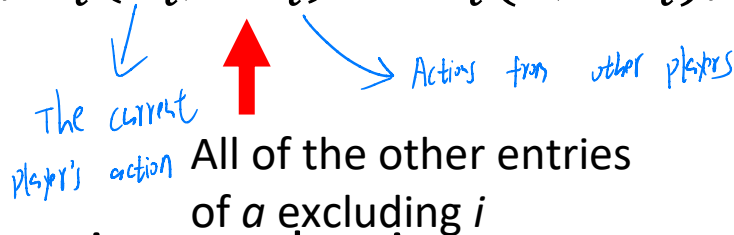
- 2 players, 2 actions: yields 2x2 payoff matrix
- Strategy set: {Stay silent, betray}

Strictly Dominant Strategies

Let's analyze such games. Some strategies are better than others!

- ★ Strictly dominant strategy: if a_i strictly better than b regardless of what other players do, a_i is strictly dominant

- I.e., $u_i(a_i, a_{-i}) > u_i(b, a_{-i}), \forall b \neq a_i, \forall a_{-i}$


The current player's action All of the other entries of a excluding i Actions from other players

(★ No matter what actions others take, a_i always the best)

- Sometimes a dominant strategy does not exist!

Strictly Dominant Strategies Example

Back to Prisoner's Dilemma

- Examine all the entries: betray strictly dominates
- Check:

Player 2		
Player 1	<i>Stay silent</i>	<i>Betray</i>
	<i>Stay silent</i>	-1, -1 -3, 0
	<i>Betray</i>	0, -3 -2, -2

Dominant →

0 > -1 -2 > -3

Dominant Strategy Equilibrium

a^* is a (strictly) dominant strategy equilibrium (DSE),
if every player i has a strictly dominant strategy a_i^*

(stable situation, in which there is no good reason for any player to change what they are doing now)

- Rational players will play at DSE, if one exists.

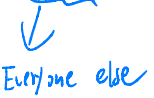
$$a^* = (a_1^*, a_2^*, a_3^*, \dots)$$

Player 1	Player 2	
	Stay silent	Betray
Stay silent	-1, -1	-3, 0
Betray	0, -3	-2, -2

In terms of single player,
the best
not good overall

Dominant Strategy: Absolute Best Responses

Player i 's best response to strategy to a_{-i} $BR(a_{-i}) = \underset{b}{\operatorname{argmax}} u_i(b, a_{-i})$

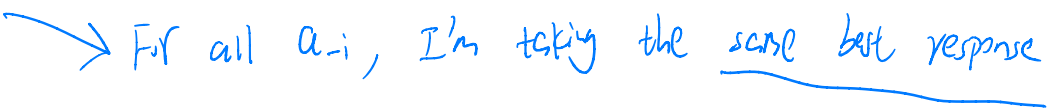
Everyone else

$BR(\text{player2=silent}) = \text{betray}$

$BR(\text{player2=betray}) = \text{betray}$

Player 2		
Player 1	<i>Stay silent</i>	<i>Betray</i>
	<i>Stay silent</i>	<i>Betray</i>
<i>Stay silent</i>	-1, -1	-3, 0
<i>Betray</i>	0, -3	-2, -2

★ a_i^* is the dominant strategy for player i , if

$a_i^* = BR(a_{-i}), \forall a_{-i}$ For all a_{-i} , I'm taking the score best response

Dominant Strategy Equilibrium

Dominant Strategy Equilibrium does not always exist.

		Player 2	
		<i>L</i>	<i>R</i>
Player 1	<i>T</i>	✓ 2, 1	0, 0
	<i>B</i>	0, 0	✓ 1, 2

↓
Depends on player 2 : No dominant strategy here

Nash Equilibrium

a^* is a Nash equilibrium if no player has an incentive to **unilaterally deviate**

$$u_i(a_i^*, \underline{a_{-i}^*}) \geq u_i(a_i, \underline{a_{-i}^*}) \quad \forall a_i \in A_i$$

↓
★ Different than the dominant strategy:
This action has to be better only if the
other player change actions / If everybody stays
the same, I also stay the same)

↓ We don't have $\forall a_{-i}$ here!

		Player 2	
		L	R
Player 1	T	2, 1	0, 0
	B	0, 0	1, 2

Nash Equilibrium: Best Response to Each Other

a^* is a Nash equilibrium:

$$\forall i, \forall b \in A_i: u_i(a_i^*, a_{-i}^*) \geq u_i(b, a_{-i}^*)$$

(no player has an incentive to **unilaterally deviate**)

★ Equivalently, for each player i :

$$\underline{a_i^* \in BR(a_{-i}^*) = \operatorname{argmax}_b u_i(b, a_{-i}^*)}$$

★ Compared to DSE (a DSE is a NE, the other direction is generally not true):

$$a_i^* = BR(a_{-i}), \forall a_{-i}$$

Nash Equilibrium: Best Response to Each Other

a^* is a Nash equilibrium:

$$\forall i, \forall b \in A_i: u_i(a_i^*, a_{-i}^*) \geq u_i(b, a_{-i}^*)$$

(no player has an incentive to **unilaterally deviate**)

- Pure Nash equilibrium:

- A **pure strategy** is a deterministic choice (no randomness).
- Later: we will consider **mixed** strategies
- In pure Nash equilibrium, players can only play pure strategies.

Finding (pure) Nash Equilibria by hand

- As player 1: For each column, find the best response, underscore it.

Player 2		
Player 1	<i>L</i>	<i>R</i>
<i>T</i>	<u>2, 1</u>	0, 0
<i>B</i>	0, 0	<u>1, 2</u>

Finding (pure) Nash Equilibria by hand

- As player 2: For each row, find the best response, upper-score it.

Player 1	Player 2	
	L	R
T	<u>2, 1</u>	0, 0
B	0, 0	<u>1, 2</u>

Finding (pure) Nash Equilibria by hand

- Entries with both lower and upper bars are pure NEs.

Mutual $\rightarrow$ Nash Equilibria (If not, then not a NE)

Player 2		L	R
Player 1	T	<u>2, 1</u>	0, 0
	B	0, 0	<u>1, 2</u>

Pure Nash Equilibrium may not exist

So far, pure strategy: each player picks a deterministic strategy. But:

Player 1	Player 2		
	<i>rock</i>	<i>paper</i>	<i>scissors</i>
<i>rock</i>	0, 0	<u>-1, 1</u>	<u>1, -1</u>
<i>paper</i>	<u>1, -1</u>	0, 0	-1, 1
<i>scissors</i>	<u>-1, 1</u>	<u>1, -1</u>	0, 0

Mixed Strategies

Can also randomize actions: “**mixed**”

- Player i assigns probabilities x_i to each action

eg: $x_i = \begin{bmatrix} \frac{1}{3} \\ \frac{1}{3} \\ \frac{1}{3} \end{bmatrix}$

$$x_i(a_i), \text{ where } \sum_{a_i \in A_i} x_i(a_i) = 1, x_i(a_i) \geq 0$$

- Now consider expected rewards

$$u_i(x_i, x_{-i}) = E_{a_i \sim x_i, a_{-i} \sim x_{-i}} u_i(a_i, a_{-i}) = \sum_{a_i} \sum_{a_{-i}} x_i(a_i) x_{-i}(a_{-i}) u_i(a_i, a_{-i})$$

Mixed Strategy Nash Equilibrium

Consider the mixed strategy $x^* = (x_1^*, \dots, x_n^*)$

- This is a **Nash equilibrium** if

probability distribution at i

$$u_i(x_i^*, x_{-i}^*) \geq u_i(x_i, x_{-i}^*) \quad \forall x_i \in \Delta_{A_i}, \forall i \in \{1, \dots, n\}$$

★ The payoffs must be equal for the actions being mixed
(not the probability distribution)

Better than doing
anything else,
“best response”

Space of probability
distributions over
strategies.

- Intuition: nobody can increase expected reward by changing only their own strategy.

Mixed Strategy Nash Equilibrium

Example: $x_1^*(\cdot) = x_2^*(\cdot) = \left(\frac{1}{3}, \frac{1}{3}, \frac{1}{3}\right)$

Player 1 \ Player 2			
	<i>rock</i>	<i>paper</i>	<i>scissors</i>
<i>rock</i>	0, 0	-1, 1	1, -1
<i>paper</i>	1, -1	0, 0	-1, 1
<i>scissors</i>	-1, 1	1, -1	0, 0

Finding Mixed NE in 2-Player 2-action Zero-Sum Game

Example: Two Finger Morra. Show 1 or 2 fingers. The “even player” wins if the sum is even, and vice versa.

	odd		
		$f1$	$f2$
even			
	$f1$	2, -2	-3, 3
	$f2$	-3, 3	4, -4

Finding Mixed NE in 2-Player 2-action Zero-Sum Game

Two Finger Morra. Two-player zero-sum game. No pure NE:

	odd		
		$f1$	$f2$
even			
	$f1$	<u>2, -2</u>	-3, 3
	$f2$	<u>-3, 3</u>	<u>4, -4</u>

Finding Mixed NE in 2-Player 2-action Zero-Sum Game

Suppose odd's mixed strategy at NE is $(q, 1-q)$, and even's $(p, 1-p)$

By definition, p is best response to q : $u_1(p, q) \geq u_1(p', q) \forall p'$.

Note $u_1(p, q) = pu_1(f_1, q) + (1 - p)u_1(f_2, q)$

Average is no greater than components

→ $u_1(p, q) = u_1(f_1, q) = u_1(f_2, q)$

		q	1-q
even	odd	f1	f2
	odd	f1	f2
p	f1	<u>2, -2</u>	-3, 3
1-p	f2	<u>-3, 3</u>	<u>4, -4</u>

Finding Mixed NE in 2-Player 2-action Zero-Sum Game

$$u_1(f_1, q) = u_1(f_2, q)$$

$$2q + (-3)(1 - q) = (-3)q + 4(1 - q)$$

$$q = \frac{7}{12}$$

$$\text{Similarly, } u_2(p, f_1) = u_2(p, f_2)$$

$$p = \frac{7}{12}$$

At this NE, even gets $-1/12$, odd gets $1/12$.

		q	1-q
		f1	f2
p	even	<u>2, -2</u>	<u>-3, 3</u>
	odd	<u>-3, 3</u>	<u>4, -4</u>

Properties of Nash Equilibrium

Major result: (John Nash '51)

- Every **finite** (players, actions) game has at least one Nash equilibrium
 - But not necessarily **pure** (i.e., deterministic strategy)
- Could be more than one
- Searching for Nash equilibria: computationally **hard**.
 - Exception: two-player zero-sum games (can be found with linear programming).

Break & Quiz

Q 2.1: Which of the following is **false**?

- (i) Rock/paper/scissors has a dominant pure strategy
- (ii) There is a Nash equilibrium for rock/paper/scissors

- A. Neither
- B. (i) but not (ii)
- C. (ii) but not (i)
- D. Both

Break & Quiz

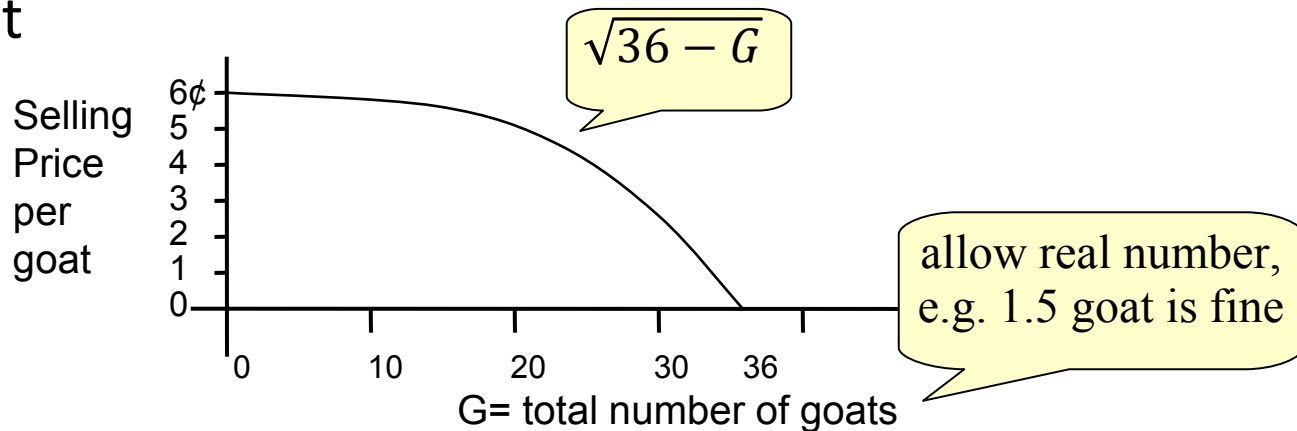
Q 2.2: Which of the following is **true**?

- (i) Nash equilibria require each player to know other players' possible strategies
- (ii) Nash equilibria require rational play

- A. Neither
- B. (i) but not (ii)
- C. (ii) but not (i)
- D. Both

Pure NE in an Infinite game: The tragedy of the Commons

- Price per goat



- How many goats should one (out of n) rational farmer graze?
- How much would the farmer earn?

Continuous Action Game

- Each farmer has infinite number of strategies $g_i \in [0, 36]$
- The value for farmer i , when the n farmers play at $(g_1, g_2, \dots, g_n)$ is

$$u_i(g_1, g_2, \dots, g_n) = g_i \sqrt{36 - \sum_{j \in [n]} g_j}$$

- **Assume** a pure Nash equilibrium exists.
- **Assume** (by apparent symmetry) the NE is $(g^*, g^*, \dots, g^*)$.

Finding g^*

- $u_i(g_1, g_2, \dots, g_n) = g_i \sqrt{36 - \sum_j g_j}$
- g^* is the best response to others ($g^*, \dots, g^*$)

$$\begin{aligned} g^* &= \operatorname{argmax}_{h \in [0, 36]} u_i(g^*, \dots, h, \dots, g^*) \\ &= \operatorname{argmax}_h h \sqrt{36 - (n-1)g^* - h} \end{aligned}$$

i-th argument



Finding g^*

$$g^* = \operatorname{argmax}_h h \sqrt{36 - (n-1)g^* - h}$$

- Taking derivative w.r.t. h of the RHS, setting to 0:

$$g^* = \frac{72 - 2(n-1)g^*}{3}$$

$$g^* = \frac{72}{2n+1} \quad \text{So what?}$$

The tragedy of the Commons

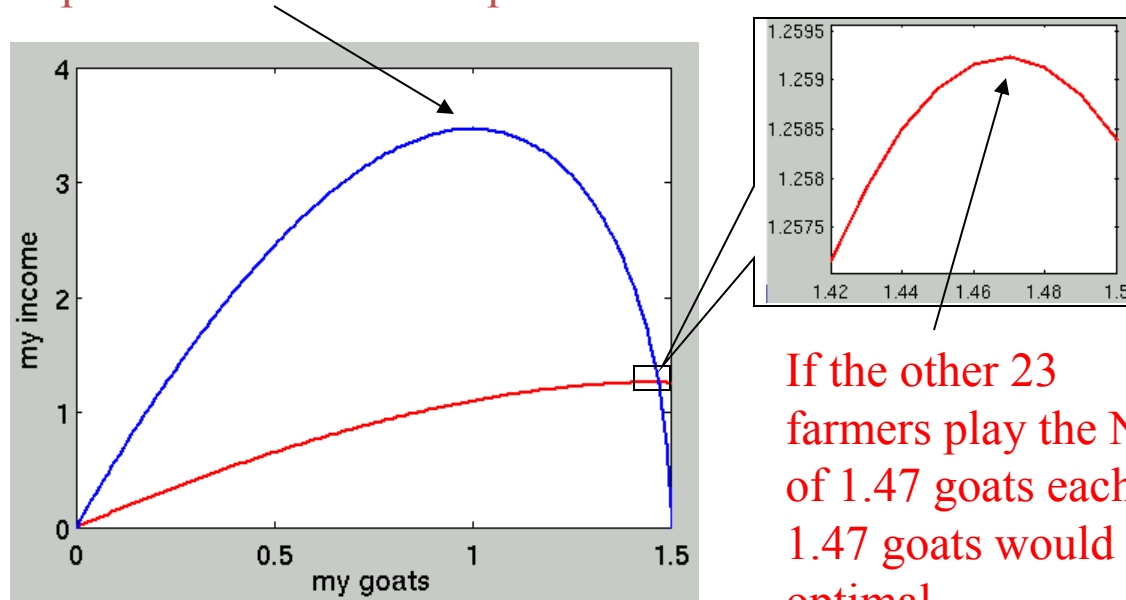
- Say there are $n=24$ farmers.
Each would **rationally** graze $g_i^* = 72/(2*24+1) = 1.47$ goats
- Each would get **1.25¢**
- But if they cooperate and each grazes only 1 goat
- Each would get **3.46¢**

(★ 纳什均衡是针对每个个体在某个假设下“所认为的利润最大化”，而不是整个系统的真实最优解)

→ this example implies the concept of "The tragedy of the commons": Overuse of the public resources to pursue single's max benefit until they destroy themselves at the end.

The tragedy of the Commons

If all 24 farmers agree on the same number of goats to raise, 1 goat per farmer would be optimal



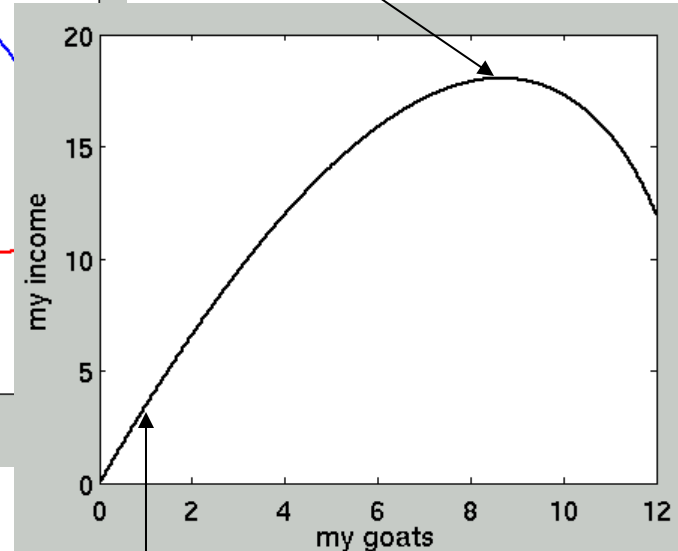
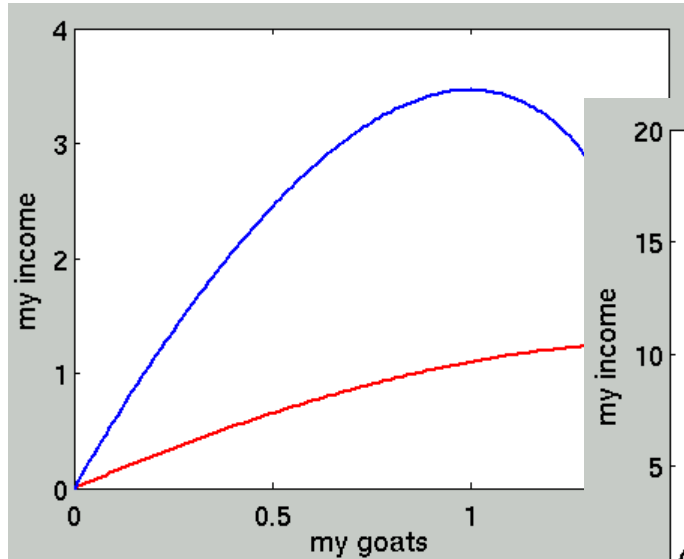
If the other 23 farmers play the N.E. of 1.47 goats each, 1.47 goats would be optimal

The tragedy of the Commons

If all 24 farmers agree on the same number of goats to raise, 1 goat per farmer would be optimal



But this is not a N.E.! A farmer can benefit from cheating (other 23 play at 1):



'by rule'

The tragedy

- Rational behaviors lead to sub-optimal solutions!
 - Maximizing individual welfare not necessarily maximizes social welfare
 - What went wrong?
-

Shouldn't have allowed **free** grazing?

It's not just the  : the use of the atmosphere and the oceans for dumping of pollutants.

Mechanism design: designing the rules of a game

Summary

- Intro to game theory
 - Characterize games by various properties
- Mathematical formulation for simultaneous games
 - Normal form, dominance, Nash equilibria, mixed vs pure